Pin No.	Designation	Description of Function
9	RESET	RESET input. A "1" level applied to this pin clears all flag and status flip-flops and forces the program counter to 0. To completely clear all of the address and index registers, RESET must be applied for 64 clock cycles (8 machine cycles).
10	TEST	TEST input. The logical state of this input can be examined with the JCN instruction.
11	CM-ROM	This pin enables a ROM bank, which can contain up to 4K words of program using 4001, 4308, or 4289. It also enables the GP I/O devices (4207, 4209, 4211) which are attached to the CM-ROM.
12	V_{DD}	Main supply voltage to the processor. Value must be V_{SS} -15.0V ±5%.
13-16	CM-RAM_0-CM-RAM_3	CM-RAM outputs. These outputs act as bank select signals for the 4002 RAM chips in the system.

Basic Timing

The MCS 40 system requires two non-overlapping clock phases, ϕ_1, ϕ_2 which are supplied by the 4201 clock generator. The 4004 will generate a SYNC signal every 8 clock periods and will send it to the other system components. The SYNC signal marks the beginning of each instruction cycle. The other MCS-40™ components will use the SYNC, ϕ_1, and ϕ_2 signals to generate external timing.

A typical machine cycle starts with the CPU sending a synchronization signal (SYNC) to the ROM's and RAM's. Next, 12 bits of ROM address are sent to the data bus using three clock cycles (@ .75 MHz). The address is then incremented by one and stored in the program counter. The selected ROM sends back 8 bits of instruction or data during the following 2 clock cycles. This information is stored in two registers: OPR and OPA. The next three clock cycles are used to execute the instruction. (See Basic Instruction Cycle.)

Basic Description of Major Circuit Blocks

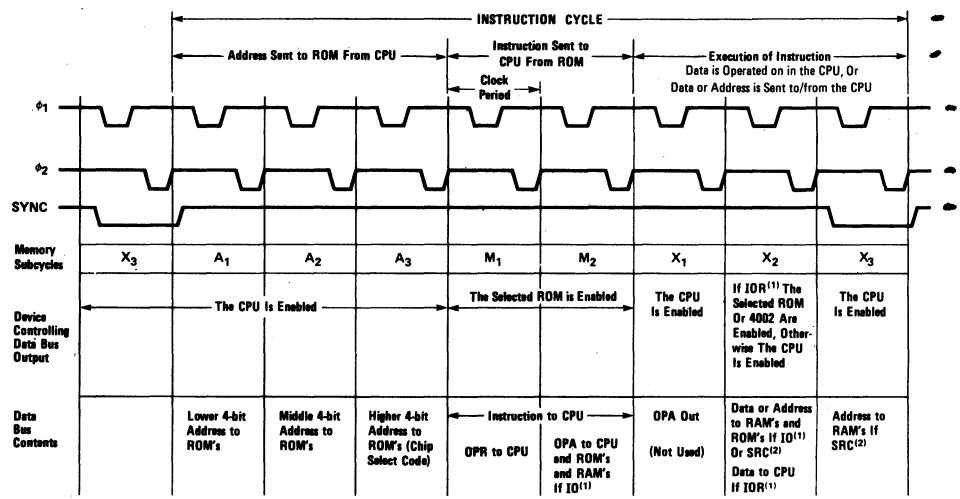
The 4004 block diagram shown contains the following functional blocks:

1. Address register (program counter and stack organized as 4 words of 12 bits each) and address incrementer.
2. Index register (64 bits organized as 16 words of 4 bits each.
3. 4 bit adder.
4. Instruction register (8 bits wide), decoder and control.
5. Peripheral circuitry.

The functional blocks communicate internally through a 4-line bus. The function and composition of each block is as follows:

1. Address Register (Program Counter & Stack) & Address Incrementer

The address register is a dynamic RAM cell array of 4 x 12 bits. It contains one level used to store the instruction



(1) IO instructions control the flow of information between accumulator in CPU, I/O lines in ROM's and RAM's and RAM storage. IOR stands for IO Read. In this case the CPU will receive data from RAM storage locations or I/O input lines.

(2) The SRC instruction designates the chip number and address for a following IO instruction.

Figure 1-2. 4004 Basic Instruction Cycle.